

Girish Yandigeri

PROGRAM – 8

Commands

`minikube start`

Starts the Minikube Kubernetes cluster.

`mkdir program-8`

`cd program-8`

Creates and enters the working directory for the program.

`touch server.js nodejs-configmap.yaml nodejs-pod.yaml`

Creates the required files for Node.js app, ConfigMap, and Pod.

`kubectl apply -f nodejs-configmap.yaml`

Creates the ConfigMap in Kubernetes.

`kubectl apply -f nodejs-pod.yaml`

Creates the Node.js Pod using the ConfigMap.

`kubectl get pods`

Displays the current pod status.

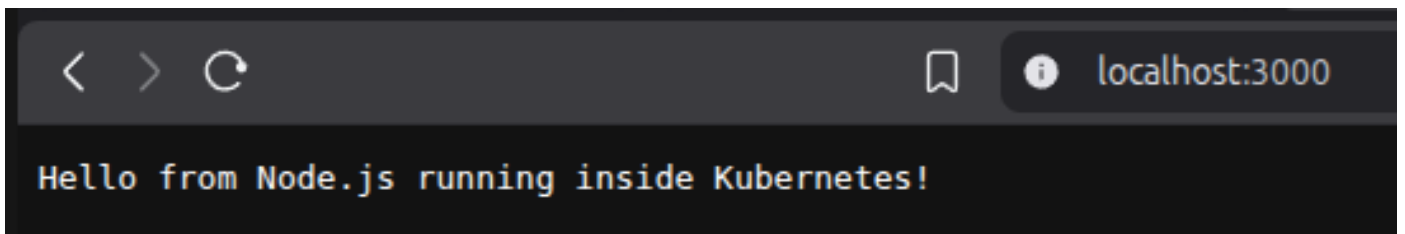
`kubectl logs nodejs-pod`

Shows the logs of the Node.js pod.

`kubectl port-forward pod/nodejs-pod 3000:3000`

Forwards port 3000 of the pod to local machine port 3000.

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ touch server.js nodejs-configmap.yaml nodejs-pod.yaml
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ nano server.js
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ nano nodejs-configmap.yaml
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ nano nodejs-pod.yaml
nodejs-pod.yaml: command not found
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ nano nodejs-pod.yaml
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-configmap unchanged
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl get pods
NAME                                READY    STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-cxfs6   1/1      Running             0           5m28s
nginx-deployment-6f9664446b-dchkr    1/1      Running             0           5m28s
nodejs-pod                           0/1      ContainerCreating   0           4s
Error from server (BadRequest): container "nodejs-container" in pod "nodejs-pod" is waiting to start: ContainerCreating
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl get pods
NAME                                READY    STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-cxfs6   1/1      Running             0           6m36s
nginx-deployment-6f9664446b-dchkr    1/1      Running             0           6m36s
nodejs-pod                           1/1      Running             0           72s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```



server.js

```
const http = require("http");

const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

nodejs-configmap.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-config
data:
  server.js: |
    const http = require("http");

    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

nodejs-pod.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
spec:
  containers:
    - name: nodejs-container
      image: node:18
      command: ["node", "/app/server.js"]
      ports:
```

- containerPort: 3000

volumeMounts:

- name: nodejs-volume
mountPath: /app

volumes:

- name: nodejs-volume

configMap:

- name: nodejs-config